

Natural Language Processing for Information Extraction and Predictive Modeling in Healthcare

by

Sonish Sivarajkumar

B.Tech in Electrical Engineering, APJ Abdul Kalam Technological University, 2020

Submitted to the Graduate Faculty of the
School of Computing and Information in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy

University of Pittsburgh

2025

UNIVERSITY OF PITTSBURGH

SCHOOL OF COMPUTING AND INFORMATION

This dissertation was presented

by

Sonish Sivarajkumar

It was defended on

July 16, 2025

and approved by

Daqing He, PhD, Professor, Department of Informatics and Networked Systems, University
of Pittsburgh

Shyam Visweswaran, MD, PhD, Professor, Department of Biomedical Informatics,
University of Pittsburgh

Yufei Huang, PhD, Professor, Department of Electrical and Computer Engineering,
University of Pittsburgh

Sunyang Fu, PhD, Assistant Professor, Department of Clinical and Health Informatics,
UTHealth Houston

Thesis Advisor/Dissertation Director: Yanshan Wang, PhD, Assistant Professor,
Intelligent Systems Program, University of Pittsburgh

Copyright © by Author's Full Name

2025

Natural Language Processing for Information Extraction and Predictive Modeling in Healthcare

Sonish Sivarajkumar

University of Pittsburgh, 2025

The integration of artificial intelligence(AI) into clinical medicine is revolutionizing healthcare by offering deeper insights from both unstructured clinical narratives and structured electronic health records (EHRs). This dissertation contributes foundational methods and insights that address immediate needs in information extraction and multimodal prediction in healthcare, emphasizing data efficiency, fairness, and robustness. The study introduces different clinical natural language processing(NLP) frameworks for adapting pretrained large language models(LLMs) to clinical information extraction via zero- and few-shot prompting achieves high-accuracy entity and relation extraction with minimal annotation. These techniques are validated in real-world hospital settings—sleep disorder notes, rehabilitation records, and lung cancer reports—where hybrid pipelines combining prompt-based, rule-based, and machine-learning components demonstrate expert-level performance despite heterogeneous documentation.

By incorporating additional data modalities, this research introduces foundational models that integrate clinical narratives with structured EHRs using transformer-based architectures to capture the contextual nuances of clinical data. These models emphasize robustness and fairness in predictive analytics, aiming to enhance the accuracy of outcome predictions and ensure equitable application across patient populations. The ultimate aim of the dissertation is to transform advancements in NLP and AI into concrete enhancements in clinical data quality, thereby improving healthcare outcomes and decision support, all while upholding the trust, safety, and equity that patients and providers rightfully expect. The work presented herein represents a

step toward that goal and lays the groundwork for academic inquiry and practical innovations in the years to come.

PREVIEW

Table of Contents

Preface.....	xiv
1.0 Introduction.....	1
1.1 Problem Statement and Motivation	1
1.2 Research Objectives	4
1.3 Dissertation Overview	7
2.0 Background and Related Work.....	9
2.1 Evolution of Clinical NLP.....	9
2.2 Comparative Studies and Benchmarking Efforts	18
2.3 Real-World Deployment Challenges for Clinical NLP	20
2.4 Integrating Unstructured Text with Structured EHR Data.....	22
2.5 Bias and Fairness in Integrated Models (Representation Learning).....	25
3.0 Adaptation of Pretrained Language Models for Clinical Information Extraction.....	29
3.1 Introduction of Clinical Information Extraction.....	29
3.2 Background	31
3.3 Literature Review	34
3.4 HealthPrompt: A Zero-shot Learning Paradigm for Clinical NLP	37
3.4.1 Materials and Methods.....	39
3.4.2 Results	42
3.4.3 Discussion.....	50
3.4.4 Conclusion.....	51
3.5 Empirical Evaluation of LLM Prompting Strategies.....	51

3.5.1 Materials and Methods	52
3.5.2 Results	60
3.5.3 Discussion.....	67
3.5.4 Conclusion.....	69
3.6 Chapter Conclusion and Discussion	70
4.0 Clinical NLP development on real world hospital settings	75
4.1 Introduction	75
4.2 Background.....	76
4.3 Literature Review	84
4.4 Study 1: Sleep Information Extraction from AD Clinical Notes	89
4.4.1 Data and Annotation.....	89
4.4.2 NLP Models Development.....	94
4.4.3 Results	98
4.4.4 Discussion.....	103
4.4.5 Study Conclusion.....	105
4.5 Study 2: Physical Rehabilitation Exercise Information Extraction.....	106
4.5.1 Ontology and Data Preparation.....	106
4.5.2 Model Development and Evaluation	112
4.5.3 Results	115
4.5.4 Discussion.....	119
4.5.5 Study Conclusion.....	123
4.6 Study 3: Lung Cancer Treatment–Response Extraction	123
4.6.1 Hybrid System Design	126

4.6.2 Data Annotation and Model Training.....	133
4.6.3 Results	136
4.6.4 Discussion.....	141
4.6.5 Study Conclusion.....	144
4.7 Chapter Conclusion and Discussion	146
5.0 Unified Representation Learning using EHRs.....	152
5.1 Introduction	152
5.2 Background.....	153
5.3 Literature Review	155
5.4 Fair Patient Model (FPM)	158
5.4.1 Methods.....	160
5.4.2 Fair Patient Model (FPM)	170
5.4.3 Results	173
5.4.4 Discussion.....	183
5.4.5 Study Conclusion.....	184
5.5 Generative Deep Patient (GDP)	185
5.5.1 Methods.....	190
5.5.2 Generative Deep Patient(GDP) Model	193
5.5.3 Results	204
5.5.4 Discussion.....	214
5.5.5 Study Conclusion.....	219
5.6 Chapter Conclusion and Discussion	220
6.0 Conclusion and Future Work	225

6.1 Key Findings and Contributions.....	225
6.2 Other Design Considerations.....	230
6.3 Limitations and Future Directions.....	232
6.4 Emerging Opportunities in Clinical NLP and AI.....	238
6.5 Conclusion.....	241
Reference List.....	243

PREVIEW

List of Tables

Table 3.1 An example of the prompt-based learning process	33
Table 3.2. Four different prompt templates used for the experiment.....	42
Table 3.3 Distribution of different phenotype categories in the MIMIC-III phenotype subset used in this study.....	43
Table 3.4 Results on the MIMIC-III phenotype subdataset with six PLMs.....	45
Table 3.5 Error Analysis Table.....	49
Table 3.6 Task Descriptions	53
Table 3.7 Evaluation data sets and samples for different tasks.....	54
Table 3.8 Performance comparison of different prompt types and language models.....	61
Table 4.1 Sleep-related keywords.....	89
Table 4.2 Sleep-related concept categories	92
Table 4.3 Prevalence of sleep-related concepts in annotated.....	93
Table 4.4 Regular expression rules.....	96
Table 4.5 Performance of sleep information extraction models by concept.....	102
Table 4.6 Ontology categories for rehabilitation exercise information with example values.	108
Table 4.7 Demographic information of patients	109
Table 4.8 Extraction performance for selected rehabilitation concepts	118
Table 4.9 Regular expression patterns for entity detection concepts.....	128
Table 4.10 Annotation schema for treatment and response entities.	134
Table 4.11 Performance of rule-based entity extraction	138

Table 4.12 Performance of rule-based entity extraction	140
Table 4.13 Summary of Chapter 4 discoveries and their practical implications.	151
Table 5.1 Comparison of accuracy for different downstream classifiers on the mortality prediction tasks using FPM.....	175
Table 5.2 Comparison of fairness and performance metrics for different models on mortality prediction tasks.	177
Table 5.3 Overall train loss, validation loss, and reconstruction loss	181
Table 5.4 The best feature reconstructions during unsupervised pre-training	182
Table 5.5 Discriminative performance metrics on EHR prediction tasks	207
Table 5.6 Automated generation metrics for discharge summary	209
Table 5.7 Automated generation metrics for discharge summary	210
Table 5.8 Ablation of Auxiliary Objectives on Discriminative Tasks.	213
Table 5.9 Insights from unified clinical representation models.....	223

List of Figures

Figure 1.1 Overall structure of the dissertation	6
Figure 3.1 Architecture of the HealthPrompt framework.	39
Figure 3.2 Chunk Encoding and Pooling.....	41
Figure 3.3 Gridsearch results: F1 score vs number of tokens in chunk encoder	46
Figure 3.4 Chunk Overlap	47
Figure 3.5 Chunk Encoding and Pooling.....	47
Figure 3.6 Ablation study of F1 score performance by chunk-encoder.....	48
Figure 3.7 Iterative prompt design process	55
Figure 3.8 Comparison of prompt type performance across five clinical NLP tasks (using GPT-3.5 in zero-shot mode)	68
Figure 4.1 Overview of the NLP information extraction pipelinede	82
Figure 4.2 Summary of the annotated corpus.....	111
Figure 4.3 Example of automated information extraction from a de-identified real-world lung cancer clinical note	125
Figure 4.4 Overview of the end-to-end NLP system for extracting treatments and RECIST responses from clinical notes.....	130
Figure 5.1 Imbalance in data across gender and mortality labels.....	164
Figure 5.2 Distribution of patients in each of the 4 tasks	168
Figure 5.3 An overview of the patient representation pipeline.....	171
Figure 5.4 Schematic of a generalist multimodal architecture for clinical foundational models.....	189

Figure 5.5 Overview of data processing and experimental pipeline.	190
Figure 5.6 Detailed GDP model components.....	194
Figure 5.7 Discriminative performance across clinical prediction tasks.	206
Figure 5.8 Human evaluation radar chart.....	211

PREVIEW

Preface

മാതാ പിതാ ഗുരു ദൈവം

To amma and achan. For all the love and support.

My journey of completing this research has been transformative and deep humbling. Raised in a small village in the southern state of Kerala, India, pursuing a Ph.D. was far beyond what I ever imagined. Coming to the United States and building an academic and research career here felt like destiny unfolding - something that happened through God's grace and the unwavering support of so many people around me. As someone who was once insecure about nearly everything, I cannot fully express how grateful and humbled I am to stand at this stage today.

When I first arrived at the University of Pittsburgh's School of Computing and Information, I aspired to bridge the gap between unstructured clinical narratives and actionable predictive models that can meaningfully improve patient care. Over the past four years, that goal guided my work - driving me to design and evaluate novel NLP and multimodal modeling techniques on real-world electronic health record (EHR) data. At every step, I was fortunate to receive guidance, encouragement, and unflagging support from many remarkable individuals.

I owe this dissertation first and foremost to my family and closest supporters. To my mother, Sudharani Sivarajkumar, and my late father, Sivarajkumar S, this work is dedicated to you - your unwavering faith, love, and countless sacrifices taught me to dream beyond what I ever imagined possible. Sreelakshmi Gopakumar, my partner and greatest champion since our school days, believe in me when I doubted myself, celebrated every small victory at my side, and lifted me through my lowest moments; without your steadfast encouragement, I would never have

reached this milestone. At home, my sister Sona Sivarajkumar and brother-in-law Sandeep Nag provided a rock-solid foundation, checking in on my frustrations and sharing in my joys with equal warmth. And to my little niece Isha, whose cheerful “unicorn” smile lit up the darkest days, thank you for reminding me what truly matters. I also remain deeply grateful to my uncles, Nandagopalan and Arul Jyothi, and my aunties, Suchithra Nandagopalan and Prasanna Arul Jyothi: your parental love, prayers, and belief in my potential reassured me that I never walked alone.

My mentors have shaped both my scientific and personal growth in profound ways. From our very first meeting, Dr. Yanshan Wang - my Ph.D. advisor who patiently guided me into becoming an independent scientist. His critical insights into clinical problems, model design, and real-world deployment challenges remain the foundation of my work, and his example of empathy and integrity has made me a better person. I am equally indebted to my dissertation committee - Drs. Shyam Visweswaran, Daqing He, Yufei Huang, and Sunyang Fu - for their invaluable feedback, rigorous questions, and high standards, which pushed me to refine both the technical rigor and clinical relevance of my research. My mentor, Dr. Shameer Khader, first lifted me from my role at IQVIA into the world of academic inquiry, believing in my abilities before I believed in myself; his early mentorship set the course for my entire Ph.D. journey. And I will always remember Dr. Jasmine E. A., who as my undergraduate professor recognized my budding interest in AI and encouraged me to enroll in my first course, planting the seeds of a lifelong passion.

Transitioning to life in the United States would have felt overwhelming without the kindness of friends who became family. Dennis and his family welcomed me as a complete stranger and guided me through my earliest days on this side of the globe, making an unfamiliar country feel like home. Dr. Balu Bhasuran and Shabeek K stood by me as steadfast friends - always ready with personal or professional support whenever I needed it.

Finally, I am grateful to everyone in Pitt NAIL Lab, past and present, whose collaborative spirit through lab meetings, brainstorming sessions, and rigorous debate over error cases and parameter choices influenced the methods described in Chapters 3, 4, and 5. To all collaborators on related projects: your expertise and real-world perspectives enriched every step of this research and made its impact possible.

As I submit this dissertation, I carry forward not only the technical skills and domain knowledge acquired along the way, but also the lessons of collaboration, friendships, perseverance, and humility. From a small village in southern India to crafting algorithms that may one day improve patient care, this journey has reaffirmed that remarkable things can happen when faith, hard work, and a supportive community come together. It is my hope that the NLP techniques and integrated models presented here will serve as a foundation for future research and ultimately lead to more equitable, effective, and data-driven healthcare for patients everywhere.

1.0 Introduction

1.1 Problem Statement and Motivation

Modern healthcare generates vast quantities of clinical data that hold enormous potential for improving patient care through the development of advanced AI-based solutions. Electronic health records (EHRs) consolidate diverse information, including laboratory results, vital signs, medication orders, and crucially, unstructured free-text notes such as physician narratives, radiology reports, and discharge summaries. These narrative documents comprise approximately 80% of all clinical data[1], meaning that many key details about a patient's condition, treatment decisions, and care progression remain "hidden" in plain text. If these rich textual data are not effectively utilized, important clinical facts may be missed; indeed, studies have shown that relying only on coded structured fields can lead to erroneous conclusions or missed patients in clinical analyses, risking oversight of nuanced insights documented in narratives.

Studies have repeatedly shown that relying only on structured data fields risks missing the nuanced insights documented in clinical narratives. For example, a patient's subtle indication of worsening dyspnea in a progress note may precede quantifiable signs such as abnormal vitals, thereby offering an opportunity for early intervention that structured data monitoring alone might miss. Ignoring textual radiology findings can lead to undetected pulmonary embolism diagnoses[2, 3]. Such oversights directly impact patient safety and the quality of care. Consequently, there is a pressing need to harness the full value of unstructured clinical texts for both retrospective analyses and prospective patient management.

Natural language processing (NLP) is the key technology that enables extraction of meaningful information from unstructured clinical text. By applying NLP techniques, free-text notes can be transformed into coded data or actionable knowledge that can be used by clinical decision-support tools or predictive models. For example, NLP has been successfully used to identify specific medical findings from radiology reports with high accuracy, achieving metrics such as an 87% positive predictive value for detecting pulmonary nodules and ~99% specificity for detecting various conditions from chest X-ray reports[4]. These examples illustrate that automated text processing can harvest critical clinical information at a scale far beyond what a manual review can accomplish. Manually reviewing thousands or millions of notes to find such patterns is nearly impossible. NLP-based systems can shift through large text corpora quickly and systematically, enabling advanced analytics to drive timely, data-informed decisions.

Despite these advances, unlocking unstructured data for improved patient care remains a challenge. Early clinical NLP solutions relied heavily on manual, rule-based methods, leveraging domain expertise to encode specific lexicons and parsing rules[5, 6]. As clinical documentation grew in volume and diversity, purely rule-based NLP became difficult to maintain, scale, and adapt to diverse clinical documentation styles. More recent data-driven methods—ranging from statistical machine learning to deep neural networks—have demonstrated improved recall and adaptability[7, 8]. The arrival of pretrained language models, notably Bidirectional Encoder Representations from Transformers (BERT)[9], BioBERT[10] and ClinicalBERT[11], has substantially boosted performance on various tasks by allowing a model to learn domain-specific representations from large corpora. These pre-trained models reduce the need for massive labeled datasets and offer a rich contextual understanding. However, they can overfit smaller datasets if

not fine-tuned properly, and introduce complexity in continuous updates as medical knowledge evolves.

Following BERT, even larger models, termed **large language models (LLMs)**, have emerged, pushing NLP capabilities to new levels. LLMs, such as OpenAI's GPT-4[12] (with tens of billions of parameters), have demonstrated the ability to perform complex language understanding and generation tasks, often with minimal task-specific training. In healthcare, researchers have begun exploring LLMs for applications, such as generating patient note summaries, answering clinical questions, and assisting in diagnostic decision-making. Remarkably, some LLMs have demonstrated near-expert performance in certain evaluations; for instance, the ChatGPT model (based on GPT-3.5) was able to perform at or near the passing threshold on the United States Medical Licensing Exam (USMLE) without any specialized fine-tuning[13]. Open-source LLMs, such as Meta's LLaMA, have further enabled the community to fine-tune large models on medical text, giving rise to domain-specific LLM variants[14].

Although LLMs can exhibit near-expert performance in controlled benchmarks, they occasionally forget some information or generate misinformation (often termed hallucination or confabulation). Their extensive parameter counts pose challenges in terms of computational overhead, privacy considerations, and direct integration into clinical workflows. Moreover, clinical data are highly heterogeneous: documentation practices vary across institutions and providers, and patient populations differ; therefore, a model trained in one setting may perform poorly in another if not carefully generalized. Models can inadvertently learn biases present in clinical data, raising concerns about fairness. For example, if certain demographic groups are underrepresented in the training data, model predictions for those groups may be less accurate, potentially exacerbating healthcare disparities[15].

Addressing these challenges is essential for harnessing AI to improve patient outcomes meaningfully. Clinicians and healthcare institutions require **robust, generalizable, and fair NLP solutions** that can uncover hidden information in free-text clinical data, **integrate smoothly with structured EHR** fields, and maintain a high accuracy and reliability over time. Moreover, many state-of-the-art models (particularly complex deep learning and LLMs) operate as “black boxes” with limited explainability, which can hinder clinician trust and acceptance. Real-world applicability also demands that NLP systems be reliable and efficient in live clinical workflows. Issues like data privacy, integration with EHR software, and maintaining accuracy over time as clinical language evolves must be addressed. These challenges have motivated ongoing research to develop more robust, generalizable, and fair NLP techniques for healthcare.

The promise of this work is substantial: by surfacing critical insights from unstructured records, advanced NLP and AI systems can enhance everything from diagnostic accuracy to risk prediction, triage, and beyond—ultimately leading to more informed care decisions, reduced clinical errors, and better patient safety on a large scale.

1.2 Research Objectives

This dissertation addresses three interconnected objectives that blend technical innovation with clinical translation, advancing clinical decision support through NLP techniques:

1. **Technical Objective : Adaptation of NLP Paradigms for Clinical Information Extraction**

- Develop robust clinical information extraction frameworks by leveraging pre-trained language models that require minimal annotated data. This approach seeks to reduce the annotation burden while maintaining a high performance.
- Systematically evaluate the efficiency of prompt-based LLM methods in clinical settings, particularly on few-shot scenarios. Assess the effectiveness of various LLMs in extracting clinically relevant information from unstructured text.

2. Application Objective : Clinical NLP development on real world hospital settings

- Demonstrate the practical utility of NLP algorithms in extracting key insights across multiple clinical subdomains, such as sleep medicine, physical rehabilitation, and lung cancer treatment.
- Developing NLP systems to solve real-world clinical problems, addressing challenges including the impact of site-specific documentation practices and data heterogeneity, while ensuring that the extraction methods enhance clinical decision-making.

3. Translational Objective : Integrated Predictive AI models using EHRs

- Develop foundational models that unify the insights derived from unstructured clinical narratives using structured EHR data. This multimodal integration aims to improve outcome predictions, such as hospital readmissions and mortality risk, by capturing the full spectrum of patient information.
- Embed fairness-aware mechanisms within the modeling framework to mitigate bias, ensuring that predictive outputs perform equitably across diverse demographic subgroups, and support patient-centered care.

Together, these objectives establish a comprehensive framework that not only advances the technical foundations of clinical NLP, but also ensures their practical and equitable translation into real-world clinical decision support systems. Figure 1.1 depicts the interconnected research objectives of this dissertation.

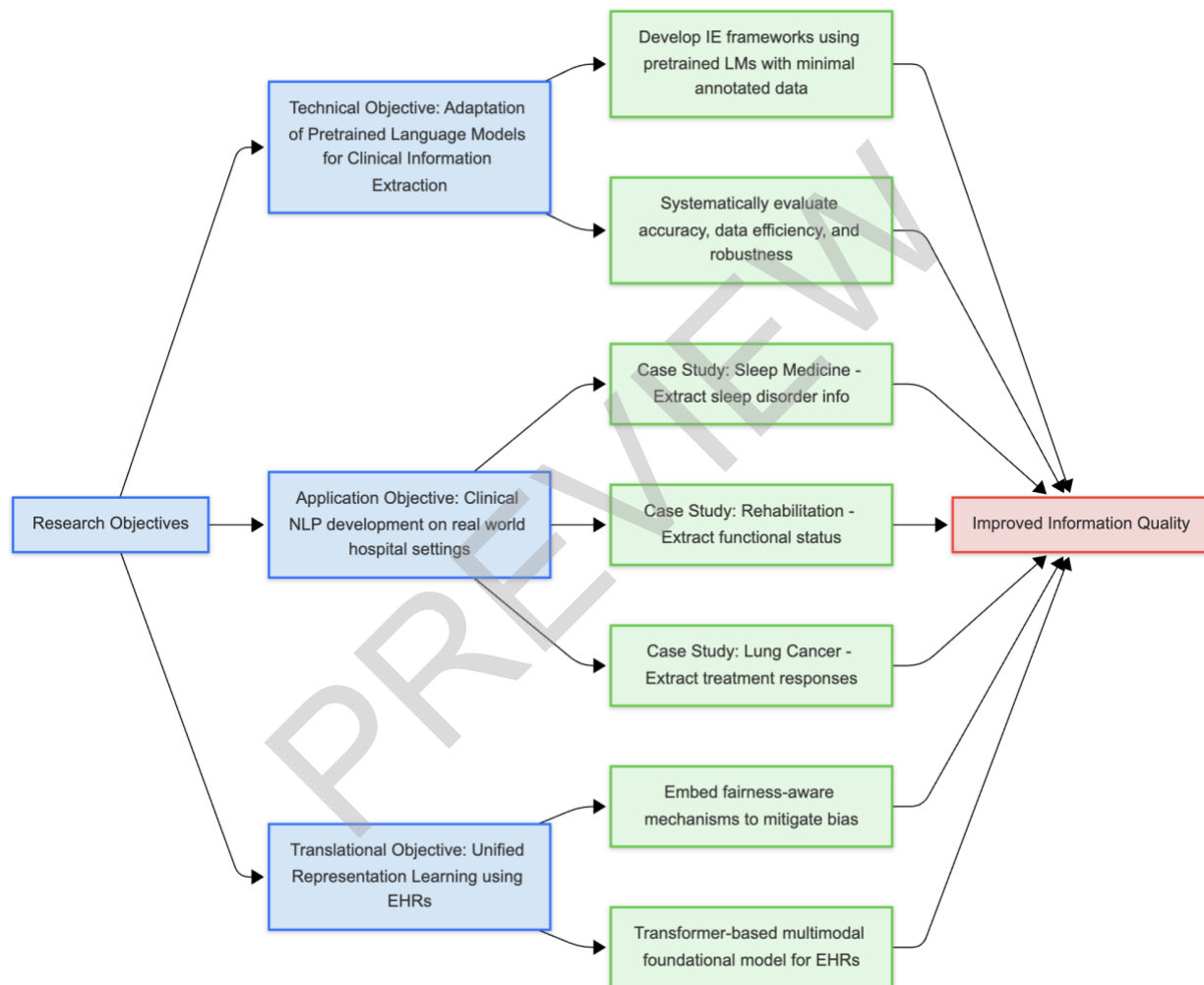


Figure 1.1 Overall structure of the dissertation

1.3 Dissertation Overview

This dissertation is organized into six chapters. Chapter 1 (the current chapter) has introduced the motivation, and objectives of this research.

Chapter 2 provides an overview of the background and literature relevant to clinical NLP in decision-support systems. It reviews different methodologies for clinical information extraction, ranging from traditional rule-based approaches to modern machine-learning and transformer-based techniques. Additionally, it discusses the challenges associated with data efficiency and generalizability in clinical NLP, and introduces the role of LLMs in addressing these limitations.

Chapter 3 presents an adaptation of LLM-driven approaches for clinical information extraction. This chapter explores the development of pretrained language model-based frameworks for extracting structured information from clinical text, particularly in settings with limited annotated data. The study also includes a detailed comparison of zero-shot and few-shot in-context learning strategies to build clinical NLP systems. The findings provide insights into the trade-offs between in-context NLP methodologies, highlighting considerations such as LLM selection and prompt development to build clinical NLP systems.

Chapter 4 focuses on the development of NLP-based information extraction models in real-world healthcare environments. It applies the best-performing NLP techniques described in Chapter 3 to clinical datasets across three distinct domains: sleep medicine, physical rehabilitation, and lung cancer treatment, and comparing that with traditional rule-based and machine learning-based methods. This chapter presents studies on extracting information about sleep disorders from narrative clinical notes and identifying functional status in physical therapy records, which are otherwise understood through manual chart review. Thirdly, this chapter presents an end-to-end NLP system-including entity extraction and relation classification - for Lung Cancer treatment-

response extraction from clinical notes, trained on UPMC datasets and externally validated at Mayo Clinic.

Chapter 5 explores the development of integrated predictive models that combine unstructured clinical narratives with structured EHRs to enhance clinical decision-making. It introduces a transformer-based foundational model designed to ingest both text-based and structured data elements (e.g., demographics, diagnoses, lab results, and vitals) for predictive modeling. This chapter describes the architecture of this model, emphasizing the techniques for multimodal fusion, temporal reasoning, and bias mitigation.

Chapter 6 synthesizes the findings from previous chapters and discusses their broader implications for clinical AI. It outlines how NLP IE techniques (Chapter 3), real-world clinical NLP development studies (Chapter 4), and integrated predictive modeling approaches (Chapter 5) collectively contribute to advancing clinical NLP research. This chapter also presents future research directions, including potential applications of foundation models in healthcare, further refinement of bias-mitigation techniques, and strategies for integrating multimodal clinical data to enhance predictive analytics.